No. of pages: 3 Registration No. ....

END SEMESTER EXAMINATIONS, AUTUMN 2025

Program: BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE & ENGINEERING)	
Subject Name: CLOUD COMPUTING	
Subject Code: 2000016102	**Semester**: VII
Duration: *2 Hours*	**Maximum Marks**: *50*

Note: Answer the following questions as directed.
Mod. No.- Module/Topic/Unit No., BL- Blooms Taxonomy, CO- Course Outcome.

Group A

(Answer All questions)
(Each question carries 2 marks)

1. **Answer the following questions:** **(5 X 2)**

	Mod. No.	BL	CO
(a) List any two advantages of cloud computing over traditional computing.	1	2	1
(b) What is a hypervisor? Name two examples.	2	1	2
(c) Mention any two features of Google App Engine.	3	1	3
(d) Name any two common cloud deployment models.	1	1	1
(e) State any two essential characteristics of cloud computing (as per NIST), other than measured service.	1	2	1

Group B

(Answer any Four questions out of Six)
(Each question carries 5 marks)

(4 X 5)

	Mod. No.	BL	CO
2. Differentiate between Grid Computing, Cluster Computing, and Cloud Computing.	1	2	1

		Mod. No.	BL	CO
3.	Explain the Infrastructure as a Service (IaaS) model with one real-world example.	1	2	1
4.	(a) Explain load balancing in virtualized environments.	2	3	2
	(b) Mention how Google Cloud implements load balancing. (2 + 3)	2	3	2
5.	Discuss the layered structure of virtualization in cloud computing.	2	2	2
6.	Write short notes on any two Google Cloud Applications.	3	2	3
7.	Explain the six stages of cloud service lifecycle management.	4	2	4

Group C

(Answer any Two questions out of Four)
(Each question carries 10 marks)

(2 X 10)

		Mod. No.	BL	CO
8.	With a neat diagram, explain the Cloud Reference Model and highlight the shift in paradigm from traditional IT.	1	4	1
9.	Analyze how Service-Oriented Architecture (SOA) and its protocol stack enable workflow and coordination across cloud services; illustrate with one PaaS example. (5 + 5)	6	4	5

Question	Mod. No.	BL	CO
10. Analyze the challenges of ensuring data security, privacy, and compliance in a public cloud setting. Provide recent, realistic examples to support the analysis. (5 + 5)	5	4	5
11. Evaluate the effectiveness of Public versus Private Cloud Storage for a large university environment, and justify which approach is better in terms of scalability, security, and cost.	6	6	6